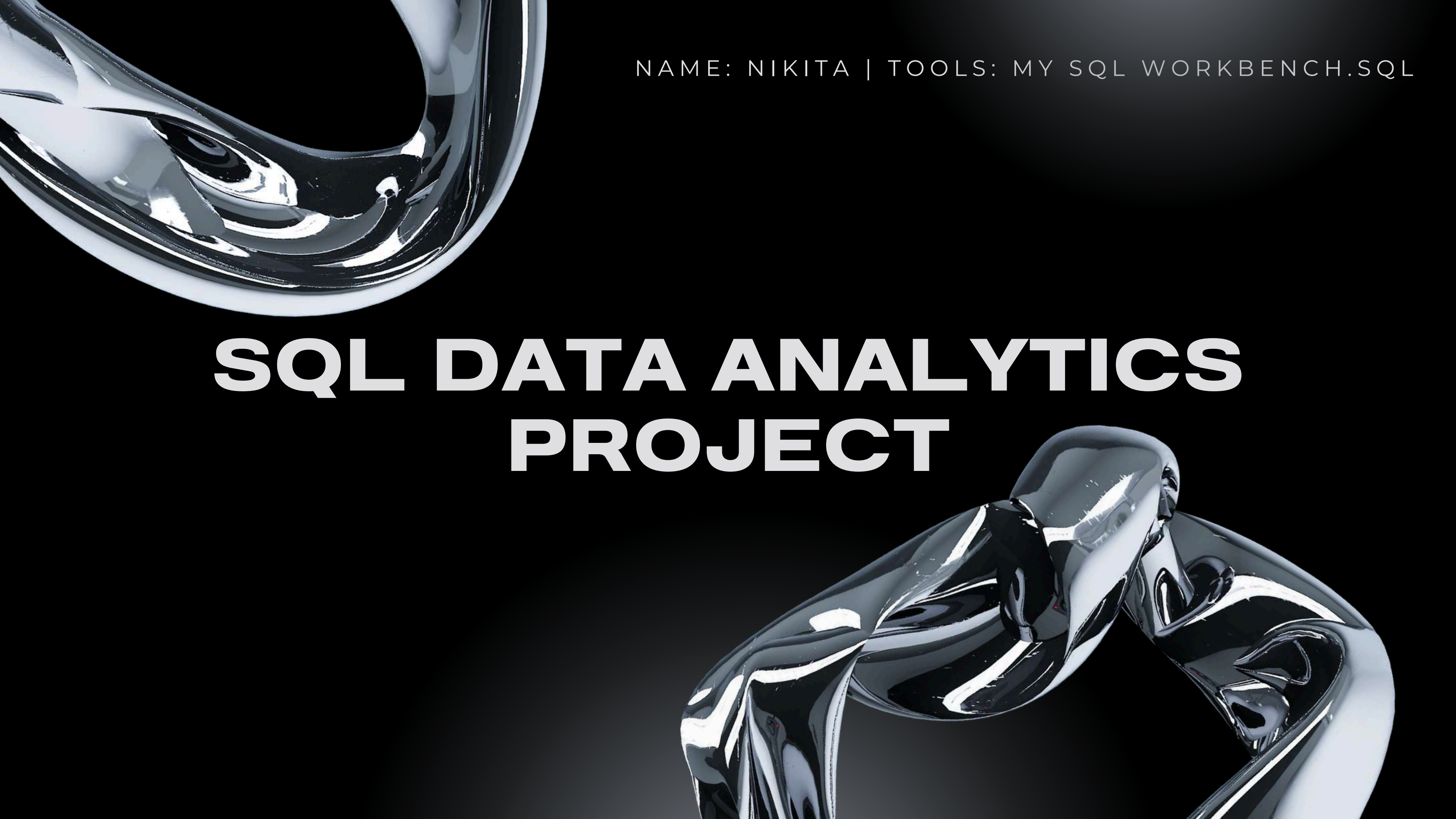




NAME: NIKITA | TOOLS: MY SQL WORKBENCH.SQL

SQL DATA ANALYTICS PROJECT



PROJECT OVERVIEW

This project analyses an urban mobility database containing interconnected data on users, drivers, vehicles, cities, rides, payments, promotions, and ratings.

Core Objective

Use SQL to convert raw mobility data into meaningful insights by examining:

- Ride activity and completion
 - Customer and driver behaviour
 - City-wise performance
 - Vehicle performance
 - Revenue and payment patterns
-

INSIDE THE URBAN MOBILITY ECOSYSTEM



PROBLEM STATEMENT & OBJECTIVES

Urban mobility platforms generate large amounts of data every day. Because this information is spread across multiple connected tables, analysing it manually can be difficult.

Key Challenges

- Managing multiple interconnected tables
- Understanding relationships between datasets
- Extracting useful information using SQL
- Comparing cities and vehicle categories
- Converting query results into actionable insights

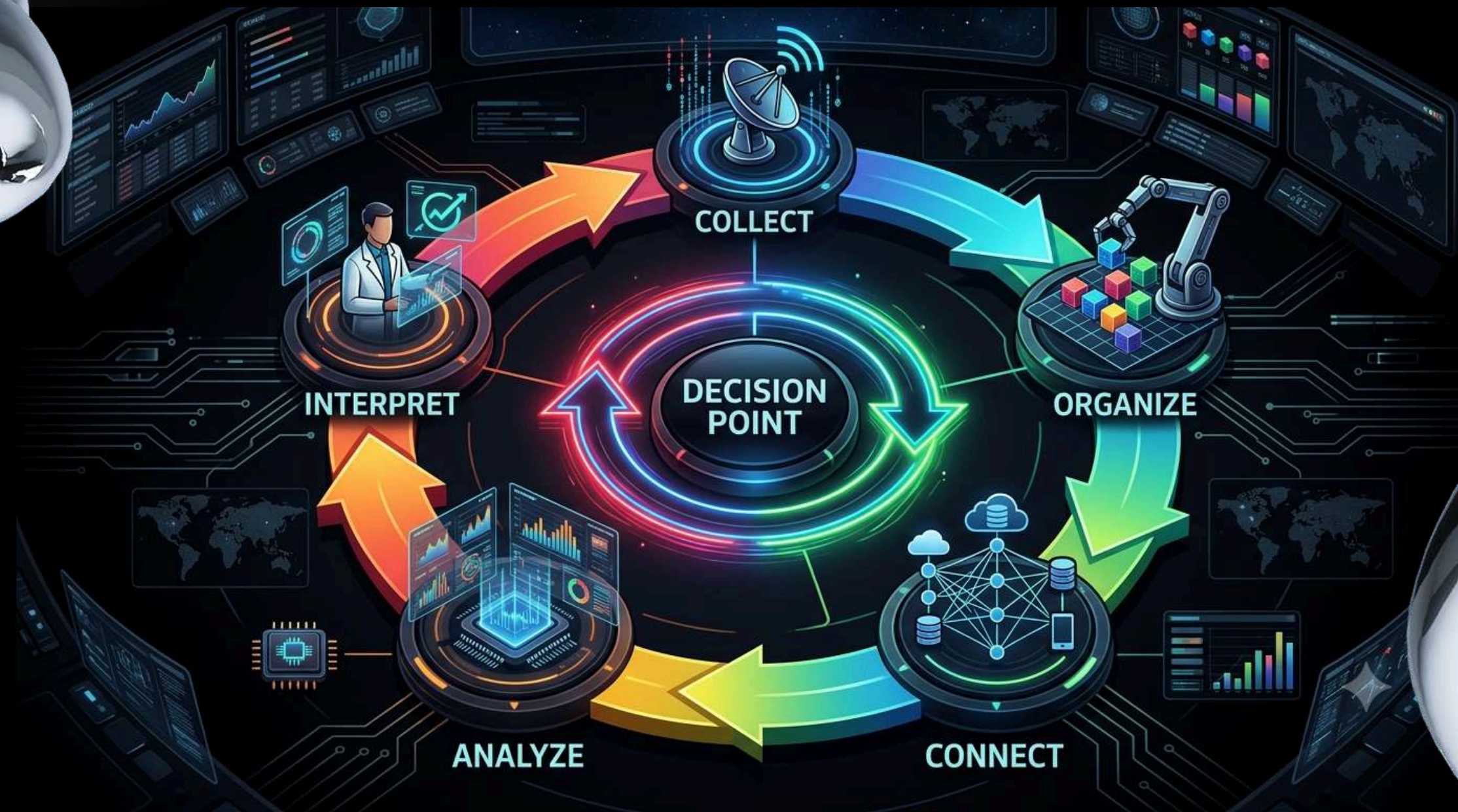
Objectives

1. Analyse ride performance
2. Understand user & driver activity
3. Compare city-wise performance.
4. Analyse revenue and payments
5. Identify important business patterns

DATA-TO-INSIGHTS PIPELINE



FROM DATA TO DECISION



SQL turns disconnected records into structured information that can support better operational decisions.

TECHNOLOGY STACK

- **MySQL**

A database is used to store and analyse mobility data.

- **MySQL Workbench**

Environment used to write and execute SQL queries.

- **Relational Database**

Multiple tables connected through keys and relationships.

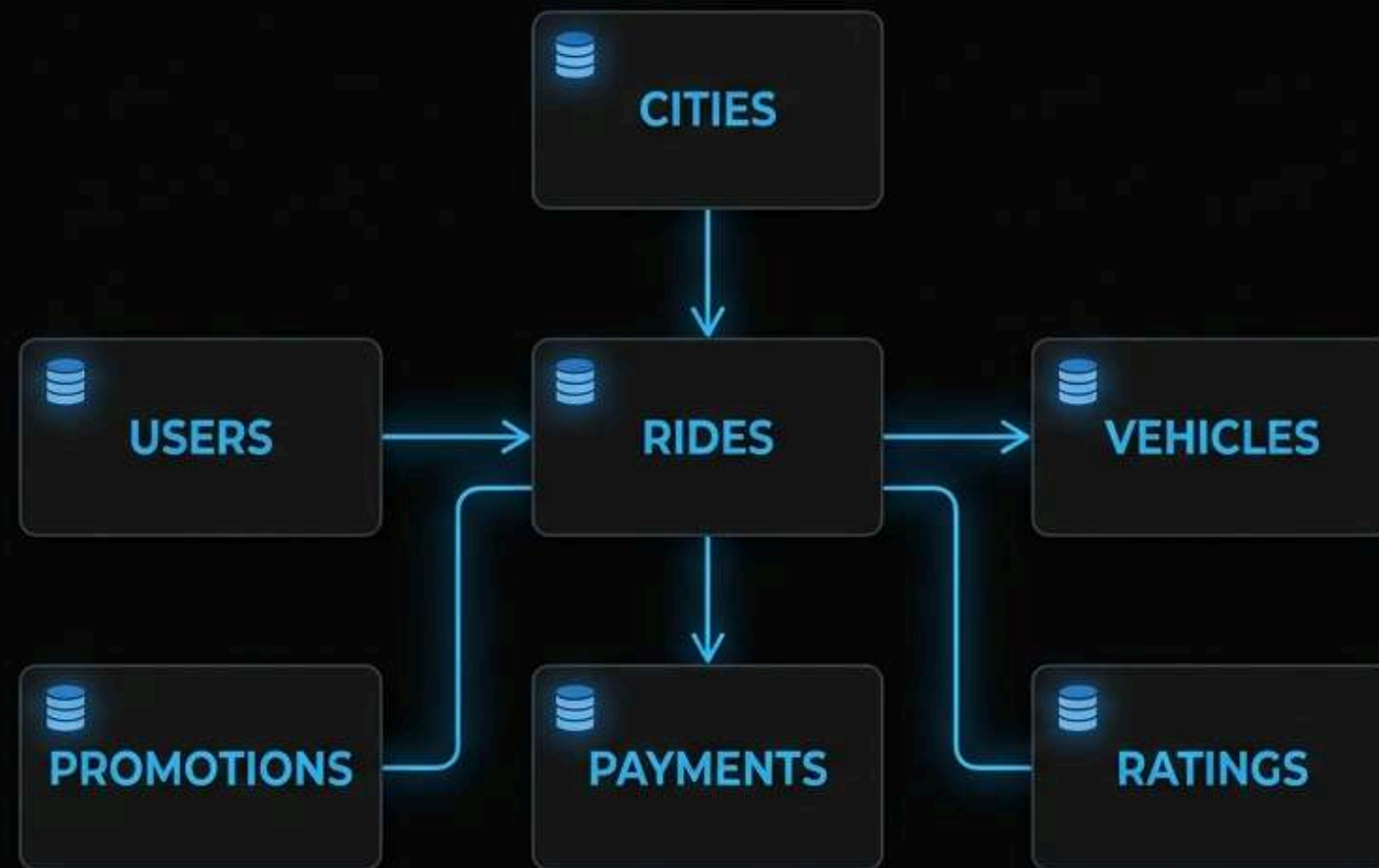
- **EER Diagram**

Used to understand tables, keys, and relationships.

- **SQL Analysis**

Used for joins, aggregation, filtering, subqueries, CTEs and calculations.

DATABASE STRUCTER



Key Tables

- Users — customer information
- Rides — trip and fare information
- Vehicles — vehicle details
- Cities — city information
- Payments — payment records
- Promotions — offers/discounts
- Ratings — ride feedback

Primary & foreign keys connect the tables and allow SQL JOINS.



SQL ANALYSIS: BASIC METRICS

```
SELECT city_name , count(ride_id) AS riders FROM rides JOIN cities  
ON rides.city_id = cities.city_id GROUP BY cities.city_id ;
```

```
select city_name , COUNT(user_id) as total_users from users join cities  
on users.city_id = cities.city_id group by city_name ;
```

These queries establish basic platform metrics and help compare user and ride activity across cities.

Query 1

Business Question:
Count how many rides in every city?

SQL Concept:
Joins, From, Group By

Query 2

Business Question:
Count how many users in every city?

SQL Concept:
Joins, From, Group By





RIDE PERFORMANCE

44,000
Total Rides

\$3.84M
Completed-Ride
Revenue

38,101
Completed Rides

CITY-WISE PERFORMANCE



Hyderabad records the highest completed-ride volume among the analysed cities.

TOP CUSTOMERS



BUSINESS QUESTION

Who are the highest-value customers?

Rank

01
02
03
04
05

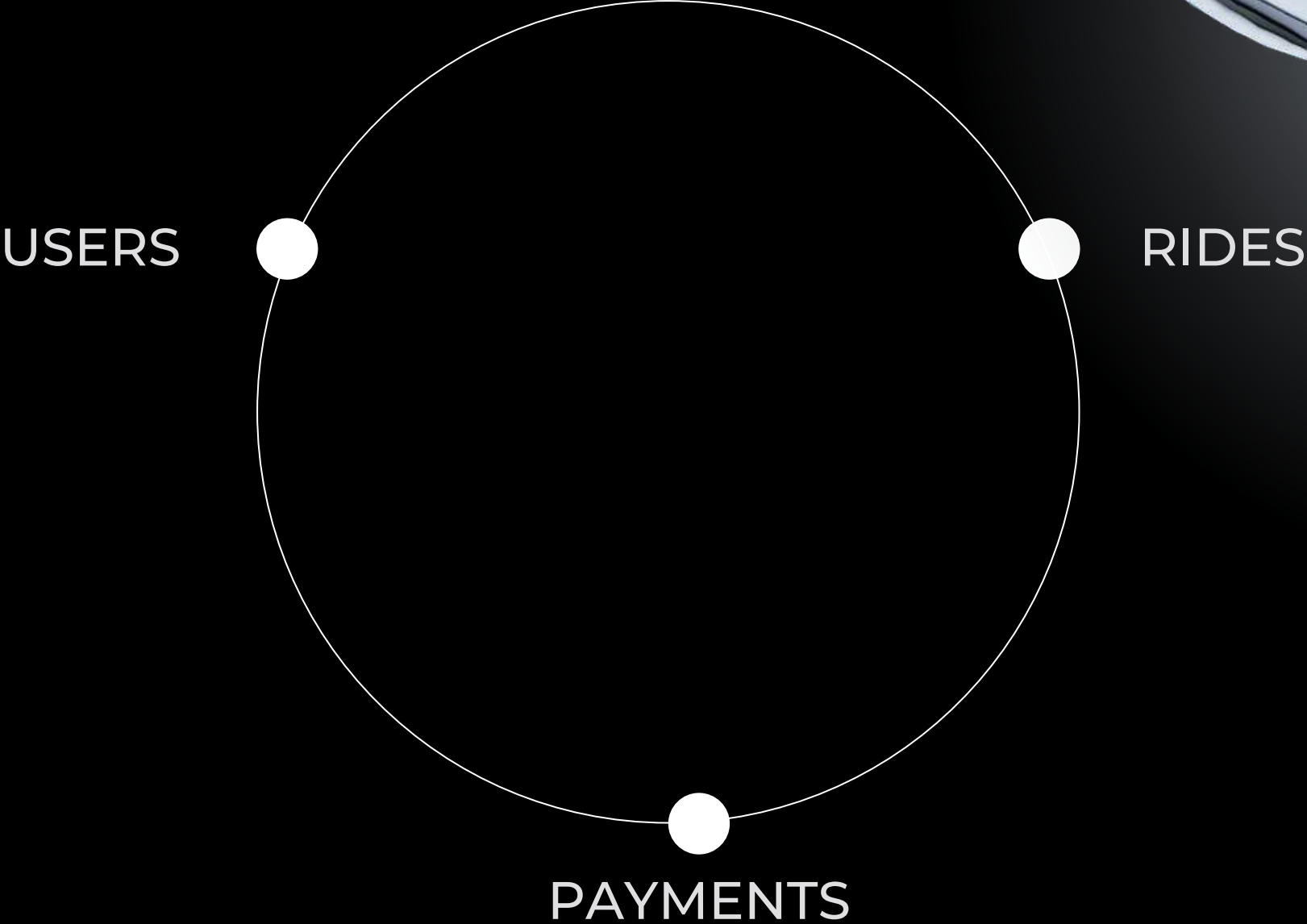
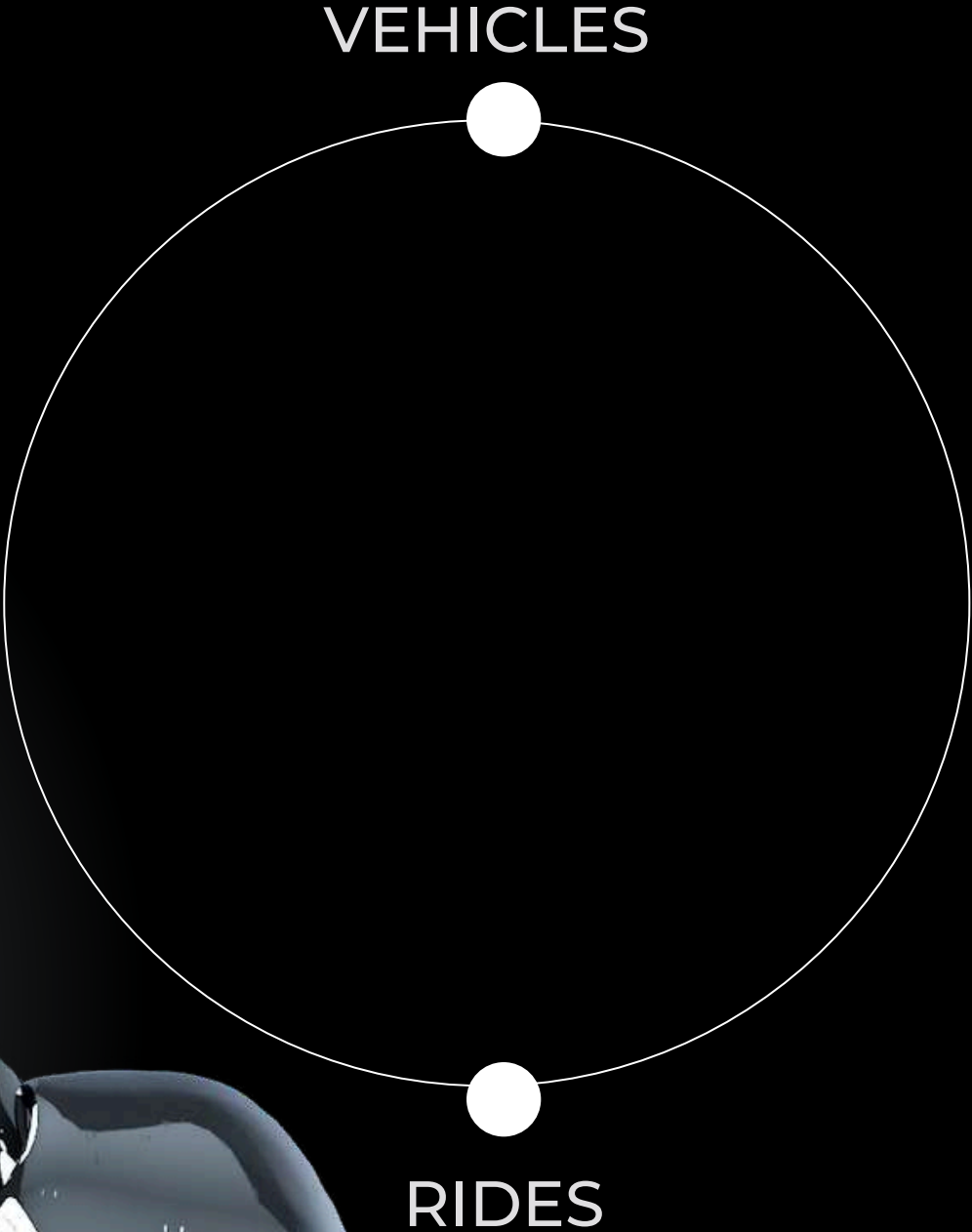
Customer

Abha Deol
Inaya Sama
Triya Subramanian
Omkaar Anne
Samuel Bhargava

Total Fare

3224.01
3089.75
3073.49
3022.53
2987.61

JOINS



```
SELECT cities.city_name , drivers.name , SUM(payments.amount) AS revenue FROM rides
JOIN drivers ON rides.driver_id= drivers.driver_id
JOIN cities ON rides.city_id=cities.city_id
JOIN payments ON rides.ride_id=payments.ride_id
GROUP BY cities.city_name,drivers.name
ORDER BY revenue DESC;
```

Why JOIN?

To combine drivers, cities and payment information for revenue analysis.



Data required for one business question is often stored in multiple tables. JOINS allow us to combine these tables using related keys.

ADVANCED SQL

CTE

Break complex analysis into steps.

WITH().

SUBQUERY

Compare values against benchmarks.

SELECT inside SELECT

WINDOW FUNCTION

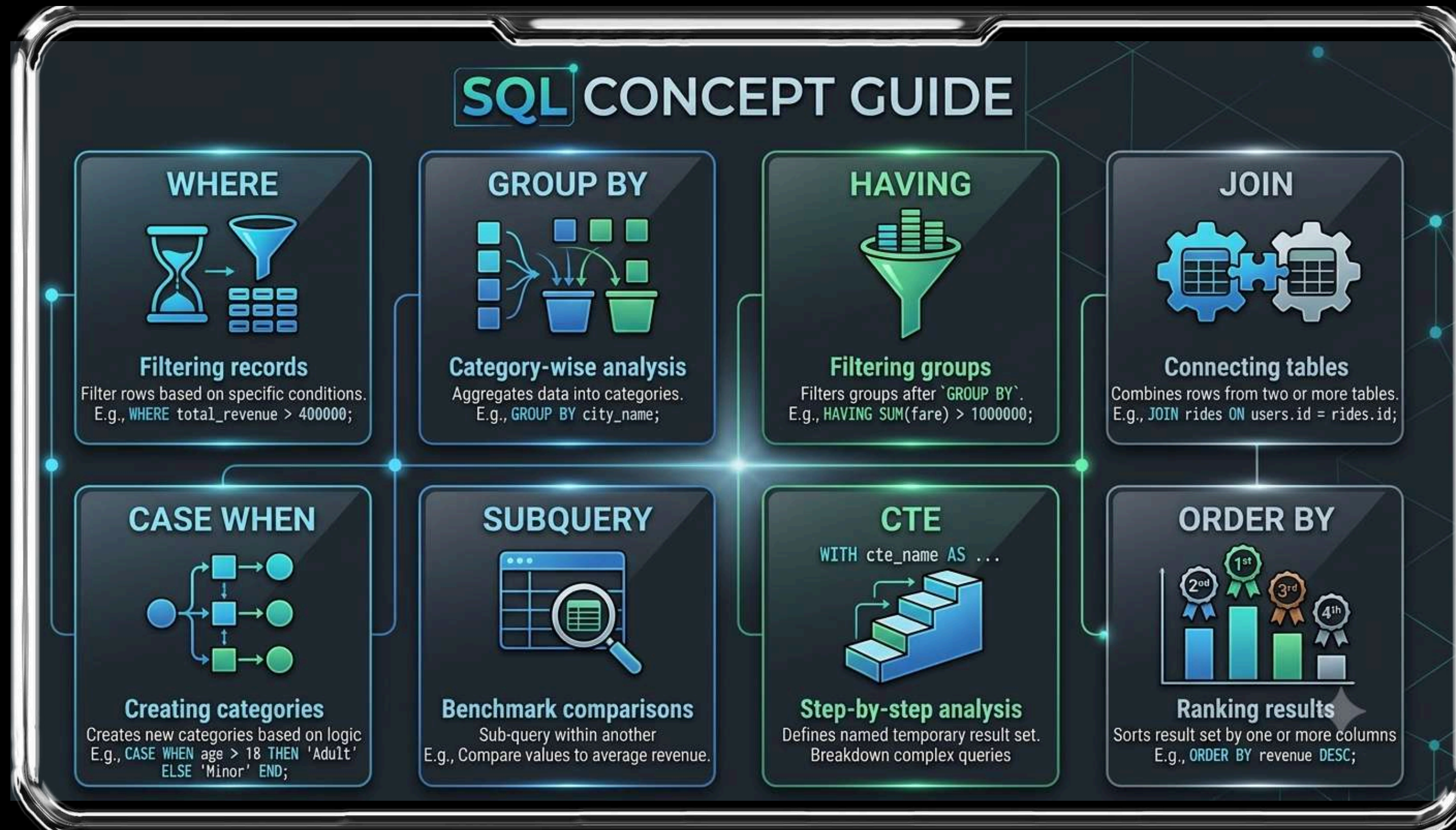
Analyse rows without collapsing the dataset.

RANK() / ROW_NUMBER().

CTE (WINDOW_CTE)

```
WITH Window_CTE AS (  
  SELECT  
    user_id,  
    city_name,  
    revenue,  
    RANK() OVER  
      (PARTITION BY city_name  
       ORDER BY revenue DESC)  
      as rev_rank  
  FROM city_revenues;  
)
```

SQL TOOLKIT USED





KEY FINDINGS: WHAT DID THE DATA TELL US?



OPERATIONS

84.67%

Ride completion rate



CUSTOMERS

3,000

Total Customers



REVENUE

\$3.84M

Completed-Ride Revenue



CITY

10

Total City



CONCLUSION & FUTURE SCOPE

The background of the slide features abstract, flowing, liquid-like shapes in shades of blue and white against a solid black background. These shapes are positioned in the corners, creating a sense of movement and depth.

Project Conclusion

- Analysed 8 interconnected mobility tables.
- Used SQL to analyse users, rides, drivers, payments, vehicles and cities.
- Applied filtering, joins, aggregation, subqueries and CASE WHEN.
- Converted raw data into business-oriented insights.



Future Scope

- Real-time mobility dashboard
- Demand forecasting
- Customer segmentation
- Driver performance monitoring
- Automated reporting



THANK YOU